Analysis of a supermart grocery sales

```
# loading the library pandas into the environment
import pandas as pd
```

```
# loading the data and saving it as 's'
s = pd.read_csv('/content/Supermart Grocery Sales - Retail Analytics Dataset.csv')
s
```

	Order ID	Customer Name	Category	Sub Category	City	Order Date	Region	Sales	Discount	Profit	State
0	OD1	Harish	Oil & Masala	Masalas	Vellore	11-08-2017	North	1254	0.12	401.28	Tamil Nadu
1	OD2	Sudha	Beverages	Health Drinks	Krishnagiri	11-08-2017	South	749	0.18	149.80	Tamil Nadu
2	OD3	Hussain	Food Grains	Atta & Flour	Perambalur	06-12-2017	West	2360	0.21	165.20	Tamil Nadu
3	OD4	Jackson	Fruits & Veggies	Fresh Vegetables	Dharmapuri	10-11-2016	South	896	0.25	89.60	Tamil Nadu
4	OD5	Ridhesh	Food Grains	Organic Staples	Ooty	10-11-2016	South	2355	0.26	918.45	Tamil Nadu
...	...	...	...	...	...	...	...	...	...	...	...
9989	OD9990	Sudeep	Eggs, Meat & Fish	Eggs	Madurai	12/24/2015	West	945	0.16	359.10	Tamil Nadu
9990	OD9991	Alan	Bakery	Biscuits	Kanyakumari	07-12-2015	West	1195	0.26	71.70	Tamil Nadu
9991	OD9992	Ravi	Food Grains	Rice	Bodi	06-06-2017	West	1567	0.16	501.44	Tamil Nadu
9992	OD9993	Peer	Oil & Masala	Spices	Pudukottai	10/16/2018	West	1659	0.15	597.24	Tamil Nadu
9993	OD9994	Ganesh	Food Grains	Atta & Flour	Tirunelveli	4/17/2018	West	1034	0.28	165.44	Tamil Nadu

9994 rows × 11 columns

```
s.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 11 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Order ID       9994 non-null   object
 1   Customer Name  9994 non-null   object
 2   Category       9994 non-null   object
 3   Sub Category   9994 non-null   object
 4   City           9994 non-null   object
 5   Order Date     9994 non-null   object
 6   Region         9994 non-null   object
 7   Sales          9994 non-null   int64
 8   Discount       9994 non-null   float64
 9   Profit         9994 non-null   float64
 10  State          9994 non-null   object
dtypes: float64(2), int64(1), object(8)
memory usage: 859.0+ KB
```

```
s.isna().sum()
```

	0
Order ID	0
Customer Name	0
Category	0
Sub Category	0
City	0
Order Date	0
Region	0
Sales	0
Discount	0
Profit	0
State	0

dtype: int64

```
len(s)
```

```
9994
```

```
s.count()
```

	0
Order ID	9994
Customer Name	9994
Category	9994
Sub Category	9994
City	9994
Order Date	9994
Region	9994
Sales	9994
Discount	9994
Profit	9994
State	9994

dtype: int64

```
s.sample(6)
```

	Order ID	Customer Name	Category	Sub Category	City	Order Date	Region	Sales	Discount	Profit	State
6877	OD6878	Verma	Beverages	Soft Drinks	Cumbum	06-07-2016	West	2202	0.20	220.20	Tamil Nadu
8500	OD8501	Ridhesh	Beverages	Soft Drinks	Ramanadhapuram	9/29/2015	West	1507	0.32	75.35	Tamil Nadu
6815	OD6816	Willams	Fruits & Veggies	Fresh Vegetables	Theni	3/13/2017	West	1400	0.24	196.00	Tamil Nadu
799	OD800	Jackson	Fruits & Veggies	Organic Vegetables	Vellore	11/27/2016	West	2018	0.34	585.22	Tamil Nadu

```
s.columns
```

```
Index(['Order ID', 'Customer Name', 'Category', 'Sub Category', 'City',
       'Order Date', 'Region', 'Sales', 'Discount', 'Profit', 'State'],
      dtype='object')
```

```
s.describe()
```

	Sales	Discount	Profit
count	9994.000000	9994.000000	9994.000000
mean	1496.596158	0.226817	374.937082
std	577.559036	0.074636	239.932881
min	500.000000	0.100000	25.250000
25%	1000.000000	0.160000	180.022500
50%	1498.000000	0.230000	320.780000
75%	1994.750000	0.290000	525.627500
max	2500.000000	0.350000	1120.950000

```
s.describe().T # T stands for transpose
```

	count	mean	std	min	25%	50%	75%	max
Sales	9994.0	1496.596158	577.559036	500.00	1000.0000	1498.00	1994.7500	2500.00
Discount	9994.0	0.226817	0.074636	0.10	0.1600	0.23	0.2900	0.35
Profit	9994.0	374.937082	239.932881	25.25	180.0225	320.78	525.6275	1120.95

```
s.describe(include="all")
```

	Order ID	Customer Name	Category	Sub Category	City	Order Date	Region	Sales	Discount	Profit	State
count	9994	9994	9994	9994	9994	9994	9994	9994.000000	9994.000000	9994.000000	9994
unique	9994	50	7	23	24	1236	5	NaN	NaN	NaN	1
top	OD9994	Amrish	Snacks	Health Drinks	Kanyakumari	09-05-2017	West	NaN	NaN	NaN	Tamil Nadu
freq	1	227	1514	719	459	38	3203	NaN	NaN	NaN	9994
mean	NaN	NaN	NaN	NaN	NaN	NaN	NaN	1496.596158	0.226817	374.937082	NaN
std	NaN	NaN	NaN	NaN	NaN	NaN	NaN	577.559036	0.074636	239.932881	NaN
min	NaN	NaN	NaN	NaN	NaN	NaN	NaN	500.000000	0.100000	25.250000	NaN
25%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	1000.000000	0.160000	180.022500	NaN
50%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	1498.000000	0.230000	320.780000	NaN
75%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	1994.750000	0.290000	525.627500	NaN

```
s["Order Date"] # .info()
```

	Order Date
0	11-08-2017
1	11-08-2017
2	06-12-2017
3	10-11-2016
4	10-11-2016
...	...
9989	12/24/2015
9990	07-12-2015
9991	06-06-2017
9992	10/16/2018
9993	4/17/2018

9994 rows × 1 columns

dtype: object

```
s["Order Date"]  = s['Order Date'].str.replace('/','-')
s["Order Date"]
```

	Order Date
0	11-08-2017
1	11-08-2017
2	06-12-2017
3	10-11-2016
4	10-11-2016
...	...
9989	12-24-2015
9990	07-12-2015
9991	06-06-2017
9992	10-16-2018
9993	4-17-2018

9994 rows × 1 columns

dtype: object

```
pd.to_datetime(s["Order Date"], format = '%m-%d-%Y')
```

	Order Date
0	2017-11-08
1	2017-11-08
2	2017-06-12
3	2016-10-11
4	2016-10-11
...	...
9989	2015-12-24
9990	2015-07-12
9991	2017-06-06
9992	2018-10-16
9993	2018-04-17

9994 rows × 1 columns

dtype: datetime64[ns]

```
s["Order Date"] = pd.to_datetime(s["Order Date"], format = '%m-%d-%Y')
```

```
s.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 11 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Order ID       9994 non-null   object
 1   Customer Name  9994 non-null   object
 2   Category       9994 non-null   object
 3   Sub Category   9994 non-null   object
 4   City           9994 non-null   object
 5   Order Date     9994 non-null   datetime64[ns]
 6   Region         9994 non-null   object
 7   Sales          9994 non-null   int64
 8   Discount       9994 non-null   float64
 9   Profit         9994 non-null   float64
 10  State          9994 non-null   object
dtypes: datetime64[ns](1), float64(2), int64(1), object(7)
memory usage: 859.0+ KB
```

```
s['Order Date'].dt.day_name()
```

	Order Date
0	Wednesday
1	Wednesday
2	Monday
3	Tuesday
4	Tuesday
...	...
9989	Thursday
9990	Sunday
9991	Tuesday
9992	Tuesday
9993	Tuesday

9994 rows × 1 columns

dtype: object

```
s['Day'] = s['Order Date'].dt.day_name()
s
```

	Order ID	Customer Name	Category	Sub Category	City	Order Date	Region	Sales	Discount	Profit	State	Day
0	OD1	Harish	Oil & Masala	Masalas	Vellore	2017-11-08	North	1254	0.12	401.28	Tamil Nadu	Wednesday
1	OD2	Sudha	Beverages	Health Drinks	Krishnagiri	2017-11-08	South	749	0.18	149.80	Tamil Nadu	Wednesday
2	OD3	Hussain	Food Grains	Atta & Flour	Perambalur	2017-06-12	West	2360	0.21	165.20	Tamil Nadu	Monday
3	OD4	Jackson	Fruits & Veggies	Fresh Vegetables	Dharmapuri	2016-10-11	South	896	0.25	89.60	Tamil Nadu	Tuesday
4	OD5	Ridhesh	Food Grains	Organic Staples	Ooty	2016-10-11	South	2355	0.26	918.45	Tamil Nadu	Tuesday
...	...	...	...	...	...	...	...	...	...	...	...	...
9989	OD9990	Sudeep	Eggs, Meat & Fish	Eggs	Madurai	2015-12-24	West	945	0.16	359.10	Tamil Nadu	Thursday
9990	OD9991	Alan	Bakery	Biscuits	Kanyakumari	2015-07-12	West	1195	0.26	71.70	Tamil Nadu	Sunday

```
s['Year'] = s['Order Date'].dt.year
```

```
s['Month'] = s['Order Date'].dt.month_name()
```

```
s
```

	Order ID	Customer Name	Category	Sub Category	City	Order Date	Region	Sales	Discount	Profit	State	Day	Year	Month
0	OD1	Harish	Oil & Masala	Masalas	Vellore	2017-11-08	North	1254	0.12	401.28	Tamil Nadu	Wednesday	2017	November
1	OD2	Sudha	Beverages	Health Drinks	Krishnagiri	2017-11-08	South	749	0.18	149.80	Tamil Nadu	Wednesday	2017	November
2	OD3	Hussain	Food Grains	Atta & Flour	Perambalur	2017-06-12	West	2360	0.21	165.20	Tamil Nadu	Monday	2017	June
3	OD4	Jackson	Fruits & Veggies	Fresh Vegetables	Dharmapuri	2016-10-11	South	896	0.25	89.60	Tamil Nadu	Tuesday	2016	October
4	OD5	Ridhesh	Food Grains	Organic Staples	Ooty	2016-10-11	South	2355	0.26	918.45	Tamil Nadu	Tuesday	2016	October
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
9989	OD9990	Sudeep	Eggs, Meat & Fish	Eggs	Madurai	2015-12-24	West	945	0.16	359.10	Tamil Nadu	Thursday	2015	December
9990	OD9991	Alan	Bakery	Biscuits	Kanyakumari	2015-07-12	West	1195	0.26	71.70	Tamil Nadu	Sunday	2015	July

```
print(s.to_string())
```

```
     Order ID Customer Name           Category        Sub Category            City Order Date   Region  Sales  Discount   Profit
0         OD1        Harish       Oil & Masala              Masalas         Vellore 2017-11-08    North   1254      0.12   401.28  Tami
1         OD2         Sudha          Beverages       Health Drinks     Krishnagiri 2017-11-08    South    749      0.18   149.80  Tami
2         OD3       Hussain        Food Grains         Atta & Flour      Perambalur 2017-06-12     West   2360      0.21   165.20  Tami
3         OD4       Jackson   Fruits & Veggies     Fresh Vegetables      Dharmapuri 2016-10-11    South    896      0.25    89.60  Tami
4         OD5       Ridhesh        Food Grains       Organic Staples            Ooty 2016-10-11    South   2355      0.26   918.45  Tami
5         OD6        Adavan        Food Grains       Organic Staples      Dharmapuri 2015-06-09     West   2305      0.26   322.70  Tami
6         OD7         Jonas   Fruits & Veggies     Fresh Vegetables          Trichy 2015-06-09     West    826      0.33   346.92  Tami
7         OD8         Hafiz   Fruits & Veggies        Fresh Fruits  Ramanadhapuram 2015-06-09     West   1847      0.32   147.76  Tami
8         OD9         Hafiz             Bakery             Biscuits     Tirunelveli 2015-06-09     West    791      0.23   181.93  Tami
9        OD10      Krithika             Bakery               Cakes         Chennai 2015-06-09     West   1795      0.27   484.65  Tami
10       OD11        Ganesh             Snacks          Chocolates           Karur 2015-06-09     West   1903      0.13   437.69  Tami
11       OD12         Yadav  Eggs, Meat & Fish               Eggs        Namakkal 2015-06-09     West    701      0.10   308.44  Tami
12       OD13        Sharon             Snacks             Cookies        Dindigul 2018-04-15    South   1659      0.19   315.21  Tami
13       OD14          Peer   Fruits & Veggies     Fresh Vegetables     Kanyakumari 2017-12-05     West   1277      0.25    63.85  Tami
14       OD15        Sundar  Eggs, Meat & Fish              Chicken     Kanyakumari 2016-11-22  Central    831      0.22   207.75  Tami
15       OD16        Ramesh       Oil & Masala   Edible Oil & Ghee     Krishnagiri 2016-11-22  Central   1440      0.11   100.80  Tami
16       OD17         Alan             Bakery               Cakes      Dharmapuri 2015-11-11  Central   1678      0.28   318.82  Tami
17       OD18        Arutra          Beverages       Health Drinks            Bodi 2015-05-13     West   1617      0.19   113.19  Tami
18       OD19       Haseena  Eggs, Meat & Fish              Mutton         Tenkasi 2015-08-27     West   1757      0.35   386.54  Tami
19       OD20         Verma          Beverages         Soft Drinks     Kanyakumari 2015-08-27     West    692      0.29   159.16  Tami
20       OD21         Hafiz          Beverages       Health Drinks         Vellore 2015-08-27     West    522      0.19   208.80  Tami
21       OD22          Alan        Food Grains       Dals & Pulses           Karur 2017-12-09  Central    948      0.13    47.40  Tami
22       OD23       Haseena          Beverages         Soft Drinks     Krishnagiri 2017-12-09  Central    707      0.34   148.47  Tami
23       OD24          Alan   Fruits & Veggies  Organic Vegetables         Tenkasi 2018-07-16     East    969      0.29    77.52  Tami
24       OD25        Sharon  Eggs, Meat & Fish               Eggs            Ooty 2016-09-25     West   1100      0.11   495.00  Tami
25       OD26      Krithika             Snacks          Chocolates     Tirunelveli 2017-01-16     West   2022      0.11   202.20  Tami
26       OD27        Muneer             Snacks             Cookies          Trichy 2017-01-16     West    541      0.12    43.28  Tami
27       OD28       Jackson             Bakery            Biscuits      Viluppuram 2016-09-17     East    979      0.22   401.39  Tami
28       OD29      Veronica          Beverages         Soft Drinks     Krishnagiri 2016-09-17     East   1988      0.22   874.72  Tami
29       OD30          Shah       Oil & Masala              Masalas     Kanyakumari 2016-09-17     East    989      0.17    69.23  Tami
30       OD31        Ramesh          Beverages         Soft Drinks      Dharmapuri 2016-09-17     East   1845      0.24   166.05  Tami
31       OD32        Mathew             Snacks              Noodles         Vellore 2016-09-17     East   1400      0.11   420.00  Tami
32       OD33         Akash       Oil & Masala   Edible Oil & Ghee          Trichy 2016-09-17     East   2163      0.26   670.53  Tami
33       OD34           Anu       Oil & Masala              Masalas         Madurai 2016-09-17     East   1689      0.34   337.80  Tami
34       OD35           Anu          Beverages       Health Drinks         Vellore 2018-10-19  Central    809      0.32    80.90  Tami
35       OD36       Sabeela  Eggs, Meat & Fish               Eggs     Tirunelveli 2017-12-08  Central   1265      0.22   569.25  Tami
36       OD37         James          Beverages         Soft Drinks           Salem 2017-12-08  Central    934      0.35    56.04  Tami
37       OD38       Willams   Fruits & Veggies  Organic Vegetables  Ramanadhapuram 2016-12-27  Central   1354      0.16   446.82  Tami
38       OD39         Jonas   Fruits & Veggies        Fresh Fruits         Chennai 2016-12-27  Central   1751      0.19   280.16  Tami
39       OD40       Sabeela  Eggs, Meat & Fish             Chicken          Cumbum 2016-12-27  Central   2045      0.23   879.35  Tami
40       OD41         Malik          Beverages       Health Drinks        Namakkal 2016-12-27  Central    660      0.19    92.40  Tami
41       OD42        Arutra          Beverages         Soft Drinks           Salem 2018-09-10  Central   1868      0.18   616.44  Tami
42       OD43        Sharon             Snacks             Cookies            Bodi 2017-07-17     West   1017      0.25   345.78  Tami
43       OD44        Mathew             Snacks             Cookies      Dharmapuri 2018-09-19    South   2311      0.20  1039.95  Tami
44       OD45        Amrish             Bakery            Biscuits     Krishnagiri 2017-03-11  Central   1663      0.17   731.72  Tami
45       OD46         Vince  Eggs, Meat & Fish               Eggs         Madurai 2017-03-11  Central   1420      0.19   355.00  Tami
46       OD47        Suresh   Fruits & Veggies  Organic Vegetables        Nagercoil 2015-10-20  Central   1415      0.31   509.40  Tami
47       OD48        Mathew             Snacks              Noodles         Chennai 2017-06-20     East   1488      0.30   610.08  Tami
48       OD49       Ridhesh             Snacks              Noodles      Pudukottai 2017-06-20     East    726      0.18    43.56  Tami
49       OD50        Amrish             Bakery               Cakes            Ooty 2016-04-18  Central   1259      0.15   415.47  Tami
50       OD51        Esther          Beverages         Soft Drinks           Salem 2016-04-18  Central    902      0.29   153.34  Tami
51       OD52          Shah          Beverages         Soft Drinks           Salem 2016-04-18  Central   2306      0.30   553.44  Tami
52       OD53         Sudha             Bakery               Cakes           Theni 2016-04-18  Central   1417      0.19    99.19  Tami
53       OD54         Yusuf   Fruits & Veggies     Fresh Vegetables         Madurai 2017-12-11     East    627      0.33   181.83  Tami
54       OD55        Esther   Fruits & Veggies      Organic Fruits        Nagercoil 2017-12-11     East    760      0.28    60.80  Tami
```

```
55        OD56        Munoon        Food Grains        Atta & Flour        Salem 2017-06-17        East        2250        0.15        611.00        Tami
```

```
pd.options.display.max_rows
```

```
60
```

```
s
```

	Order ID	Customer Name	Category	Sub Category	City	Order Date	Region	Sales	Discount	Profit	State	Day	Year	Month
0	OD1	Harish	Oil & Masala	Masalas	Vellore	2017-11-08	North	1254	0.12	401.28	Tamil Nadu	Wednesday	2017	November
1	OD2	Sudha	Beverages	Health Drinks	Krishnagiri	2017-11-08	South	749	0.18	149.80	Tamil Nadu	Wednesday	2017	November
2	OD3	Hussain	Food Grains	Atta & Flour	Perambalur	2017-06-12	West	2360	0.21	165.20	Tamil Nadu	Monday	2017	June
3	OD4	Jackson	Fruits & Veggies	Fresh Vegetables	Dharmapuri	2016-10-11	South	896	0.25	89.60	Tamil Nadu	Tuesday	2016	October
4	OD5	Ridhesh	Food Grains	Organic Staples	Ooty	2016-10-11	South	2355	0.26	918.45	Tamil Nadu	Tuesday	2016	October
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
9989	OD9990	Sudeep	Eggs, Meat & Fish	Eggs	Madurai	2015-12-24	West	945	0.16	359.10	Tamil Nadu	Thursday	2015	December
9990	OD9991	Alan	Bakery	Biscuits	Kanyakumari	2015-07-12	West	1195	0.26	71.70	Tamil Nadu	Sunday	2015	July

Data Analysis

```
s['Category'].unique()
```

```
array(['Oil & Masala', 'Beverages', 'Food Grains', 'Fruits & Veggies',
       'Bakery', 'Snacks', 'Eggs, Meat & Fish'], dtype=object)
```

```
s['Category'].nunique()
```

```
7
```

```
s['Category'].value_counts()
```

Category	count
Snacks	1514
Eggs, Meat & Fish	1490
Fruits & Veggies	1418
Bakery	1413
Beverages	1400
Food Grains	1398
Oil & Masala	1361

dtype: int64

```
for i in s.columns:
  print(i, "\n" , s[i].value_counts())
```

```
Order ID
 Order ID
OD9994    1
OD1       1
OD2       1
OD3       1
OD4       1
         ..
OD15      1
OD14      1
OD13      1
OD12      1
OD11      1
Name: count, Length: 9994, dtype: int64
Customer Name
 Customer Name
Amrish      227
Krithika    224
Verma       218
Arutra      218
Vidya       215
Shah        215
Suresh      212
Surya       209
Hussain     208
Harish      208
Sudeep      207
Komal       206
Adavan      205
Mathew      205
Veena       205
Muneer      204
Ridhesh     204
Peer        204
Veronica    203
Arvind      203
Vinne       203
Sharon      202
Haseena     202
Roshan      201
Malik       201
Yusuf       201
Ravi        200
Shree       200
Alan        198
Jonas       198
James       197
Ram         197
Amy         196
Akash       196
Rumaiza     195
Sheeba      195
Willams     195
Ganesh      193
Esther      189
Sudha       189
Ramesh      188
Sabeela     188
```

```
c = pd.crosstab(s['Category'], s['Customer Name'])
print(c.to_string)
```

```
<bound method DataFrame.to_string of Customer Name      Adavan  Aditi  Akash  Alan  Amrish  Amy  Anu  Arutra  \
Category
Bakery                 29     33     31    27      42   27   15      33
Beverages              26     27     23    23      32   33   31      30
Eggs, Meat & Fish      20     18     31    27      23   22   37      37
Food Grains            28     31     35    36      40   27   26      27
Fruits & Veggies       33     25     23    28      32   27   18      38
Oil & Masala           32     27     30    27      23   24   26      30
Snacks                 37     26     23    30      35   36   33      23

Customer Name      Arvind  Esther  ...  Surya  Veena  Verma  Veronica  Vidya  \
Category                           ...
Bakery                 33      28  ...     24     23     31        27     22
Beverages              24      25  ...     23     34     30        30     30
Eggs, Meat & Fish      26      27  ...     36     31     41        26     41
Food Grains            28      25  ...     30     23     27        29     24
Fruits & Veggies       39      25  ...     36     31     38        43     34
```

```
Oil & Masala           29      21  ...     28     29     26        23     36
Snacks                 24      38  ...     32     34     25        25     28

Customer Name      Vince  Vinne  Willams  Yadav  Yusuf
Category
Bakery                32     26       29     27     29
Beverages             27     17       36     31     21
Eggs, Meat & Fish     36     38       33     29     22
Food Grains           29     32       27     18     25
Fruits & Veggies      19     25       27     31     23
Oil & Masala          22     25       22     22     42
Snacks                23     40       21     27     39

[7 rows x 50 columns]>
```

```
s['Sales'].min()
```

```
500
```

```
s['Sales'].sum() - s['Profit'].sum()
```

```
np.float64(11209860.8)
```

```
s['COGs'] = s['Sales'] - s['Profit']
```

```
s
```

	Order ID	Customer Name	Category	Sub Category	City	Order Date	Region	Sales	Discount	Profit	State	Day	Year	Month	C
0	OD1	Harish	Oil & Masala	Masalas	Vellore	2017-11-08	North	1254	0.12	401.28	Tamil Nadu	Wednesday	2017	November	852
1	OD2	Sudha	Beverages	Health Drinks	Krishnagiri	2017-11-08	South	749	0.18	149.80	Tamil Nadu	Wednesday	2017	November	599
2	OD3	Hussain	Food Grains	Atta & Flour	Perambalur	2017-06-12	West	2360	0.21	165.20	Tamil Nadu	Monday	2017	June	2194
3	OD4	Jackson	Fruits & Veggies	Fresh Vegetables	Dharmapuri	2016-10-11	South	896	0.25	89.60	Tamil Nadu	Tuesday	2016	October	806
4	OD5	Ridhesh	Food Grains	Organic Staples	Ooty	2016-10-11	South	2355	0.26	918.45	Tamil Nadu	Tuesday	2016	October	1436
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
9989	OD9990	Sudeep	Eggs, Meat & Fish	Eggs	Madurai	2015-12-24	West	945	0.16	359.10	Tamil Nadu	Thursday	2015	December	585
9990	OD9991	Alan	Bakery	Biscuits	Kanyakumari	2015-07-12	West	1195	0.26	71.70	Tamil Nadu	Sunday	2015	July	1123
9991	OD9992	Ravi	Food Grains	Rice	Bodi	2017-06-06	West	1567	0.16	501.44	Tamil Nadu	Tuesday	2017	June	1065
9992	OD9993	Peer	Oil & Masala	Spices	Pudukottai	2018-10-16	West	1659	0.15	597.24	Tamil Nadu	Tuesday	2018	October	1061
9993	OD9994	Ganesh	Food Grains	Atta & Flour	Tirunelveli	2018-04-17	West	1034	0.28	165.44	Tamil Nadu	Tuesday	2018	April	868

```
s['Margin'] =  s['Profit']/s['Sales']
s
```

	Order ID	Customer Name	Category	Sub Category	City	Order Date	Region	Sales	Discount	Profit	State	Day	Year	Month	C
0	OD1	Harish	Oil & Masala	Masalas	Vellore	2017-11-08	North	1254	0.12	401.28	Tamil Nadu	Wednesday	2017	November	852
1	OD2	Sudha	Beverages	Health Drinks	Krishnagiri	2017-11-08	South	749	0.18	149.80	Tamil Nadu	Wednesday	2017	November	599
2	OD3	Hussain	Food Grains	Atta & Flour	Perambalur	2017-06-12	West	2360	0.21	165.20	Tamil Nadu	Monday	2017	June	2194
3	OD4	Jackson	Fruits & Veggies	Fresh Vegetables	Dharmapuri	2016-10-11	South	896	0.25	89.60	Tamil Nadu	Tuesday	2016	October	806
4	OD5	Ridhesh	Food Grains	Organic Staples	Ooty	2016-10-11	South	2355	0.26	918.45	Tamil Nadu	Tuesday	2016	October	1436
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
9989	OD9990	Sudeep	Eggs, Meat & Fish	Eggs	Madurai	2015-12-24	West	945	0.16	359.10	Tamil Nadu	Thursday	2015	December	585
9990	OD9991	Alan	Bakery	Biscuits	Kanyakumari	2015-07-12	West	1195	0.26	71.70	Tamil Nadu	Sunday	2015	July	1123
9991	OD9992	Ravi	Food Grains	Rice	Bodi	2017-06-06	West	1567	0.16	501.44	Tamil Nadu	Tuesday	2017	June	1065
9992	OD9993	Peer	Oil & Masala	Spices	Pudukottai	2018-10-16	West	1659	0.15	597.24	Tamil Nadu	Tuesday	2018	October	1061
9993	OD9994	Ganesh	Food Grains	Atta & Flour	Tirunelveli	2018-04-17	West	1034	0.28	165.44	Tamil Nadu	Tuesday	2018	April	868

```
s.duplicated().sum()
```

```
np.int64(0)
```

```
# top 10 customers
s.groupby('Customer Name')['Profit'].sum().sort_values(ascending = False).head(10)
```

Customer Name	Profit
Arutra	87572.40
Vidya	86725.64
Krithika	85633.03
Akash	82121.26
Surya	80996.85
Amrish	80191.89
Vinne	79873.12
Amy	79159.94
Harish	79107.83
Verma	79096.53

dtype: float64

Double-click (or enter) to edit

```
# profit by category
s.groupby('Category')['Profit'].sum().sort_values(ascending = False)
```

Category	Profit
Snacks	568178.85
Eggs, Meat & Fish	567357.22
Fruits & Veggies	530400.38
Food Grains	529162.64
Bakery	528521.06
Beverages	525605.76
Oil & Masala	497895.29

dtype: float64

```
s.groupby('Year')['Profit'].sum().sort_values()
```

Year	Profit
2015	752529.11
2016	797192.99
2017	953216.22
2018	1244182.88

dtype: float64

```
s.groupby('Month')[['Profit', 'Sales']].sum().sort_values(by = 'Profit', ascending= False)
```

Month	Profit	Sales
November	556048.05	2193924
December	530036.44	2088076
September	517788.56	2064266
October	309376.90	1243289
July	274594.57	1089385
March	267347.33	1053980
May	263643.40	1086920
June	263296.16	1057808
August	258912.09	1046807
April	247476.97	998453
January	142518.52	577972
February	116082.21	456102

```
s[['Sales', 'Profit']].corr() # correlation
```

	Sales	Profit
Sales	1.000000	0.605349
Profit	0.605349	1.000000

```
# direction - positive and negative

# strength - (-1 to 1) how close the number is to 1; 0.7and above - very stong

s.groupby('Region').agg({'Profit': 'sum', 'Margin': 'mean', 'COGs': 'sum'}).sort_values(by = 'Profit', ascending = False)
```

Region	Profit	Margin	COGs
West	1192004.61	0.248333	3606738.39
East	1074345.58	0.252188	3174022.42
Central	856806.84	0.248511	2611349.16
South	623562.89	0.252952	1816898.11
North	401.28	0.320000	852.72

```
s.groupby("Year")['Profit'].sum()
```

Year	Profit
2015	752529.11
2016	797192.99
2017	953216.22
2018	1244182.88

dtype: float64

```
s.groupby(["Month", "Year"])['Profit'].sum().sort_values(ascending = False)
```

Month	Year	Profit
September	**2018**	176698.83
November	**2018**	175386.85
December	**2018**	173354.37
September	**2017**	137044.30
December	**2017**	135821.97
November	**2017**	134823.71
	2015	124528.77
	2016	121308.72
December	**2016**	117760.26
October	**2018**	115026.58
September	**2016**	108852.15
December	**2015**	103099.84
September	**2015**	95193.28
May	**2018**	88614.25
June	**2018**	87801.18
July	**2018**	86654.95
March	**2018**	82793.96
May	**2017**	80991.20
August	**2018**	79288.34
July	**2017**	77383.36
April	**2018**	76978.34
June	**2017**	69709.74
October	**2017**	69375.37
August	**2017**	66637.41
March	**2015**	66572.78
October	**2015**	63465.04
	2016	61509.91
January	**2018**	60427.90
March	**2017**	59920.01
April	**2017**	59817.03

```
ym = s.groupby(['Year', 'Month']).agg({'Profit': 'sum', 'Margin': 'mean', 'COGs': 'sum'}).sort_values(by = 'Profit', ascending = False)

ym
```

March	**2016**	58060.38
July	**2016**	56951.80
May	**2016**	56158.91
August	**2015**	54424.99
June	**2016**	54424.72
July	**2015**	53604.46
April	**2015**	51679.95
June	**2015**	51360.52
February	**2018**	41157.33
May	**2015**	37879.04
February	**2017**	32683.32
January	**2015**	32594.50
	2017	29008.80

		Profit	Margin	COGs
February	**2016**	24115.62		
January	**2018**	20487.32		
2018	**September**	176698.83	0.248410	528981.17
	November	175386.85	0.257843	508023.15
	December	173354.37	0.249762	513890.63
2017	**September**	137044.30	0.246309	409683.70
	December	135821.97	0.259261	382485.03
	November	134823.71	0.245784	416991.29
2015	**November**	124528.77	0.258585	356450.23
2016	**November**	121308.72	0.253488	356411.28
	December	117760.26	0.251930	354484.74
2018	**October**	115026.58	0.261242	328871.42
2016	**September**	108852.15	0.256075	320805.85
2015	**December**	103099.84	0.257554	307179.16
	September	95193.28	0.245373	287006.72
2018	**May**	88614.25	0.243678	278796.75
	June	87801.18	0.247878	267100.82
	July	86654.95	0.261195	250437.05
	March	82793.96	0.234538	272910.04
2017	**May**	80991.20	0.243556	251532.80
2018	**August**	79288.34	0.233716	251725.66
2017	**July**	77383.36	0.246617	238147.64
2018	**April**	76978.34	0.248227	233171.66
2017	**June**	69709.74	0.244824	221744.26
	October	69375.37	0.230051	238650.63
	August	66637.41	0.261477	189451.59
2015	**March**	66572.78	0.266624	180583.22
	October	63465.04	0.258302	178260.96
2016	**October**	61509.91	0.244337	188129.09
2018	**January**	60427.90	0.251419	174311.10
2017	**March**	59920.01	0.253742	179035.99
	April	59817.03	0.240647	187955.97

```
s.pivot_table(index = 'Day', columns= "Year", values = 'Profit', aggfunc = 'sum')              #.sort_values(by = 'Profit', ascending = Fal
```

August	58561.35	0.253648	170981.65	
March	58060.58	0.272174	154103.42	
May	56158.91	0.248836	166563.09	
June	54424.72	0.267246	159863.28	
April	51679.95	0.253333	151578.05	
February	41157.33	0.243178	125109.67	
February	32683.32	0.241928	100298.68	
January	32594.50	0.260000	89902.50	
January	29008.80	0.220674	102718.20	

Year	2015	2016	2017	2018
Day				
Friday	18634.51	9325.73	171111.61	208060.83
Monday	112745.40	119350.68	161272.15	212213.49
Saturday	136161.56	162703.69	160171.75	226858.49
Sunday	125948.86	143804.63	157018.23	227777.84
Thursday	79110.52	71754.57	27237.69	31428.09
Tuesday	148958.17	158309.76	173654.87	235442.58
Wednesday	130970.09	131943.93	102749.90	102401.56

```
s.groupby(['Day', 'Year'])['Profit'].sum()     #.sort_values(ascending = False)
```

2016	**February**	24119.62	0.260469	66707.38
	January	20487.32	0.233276	68521.68

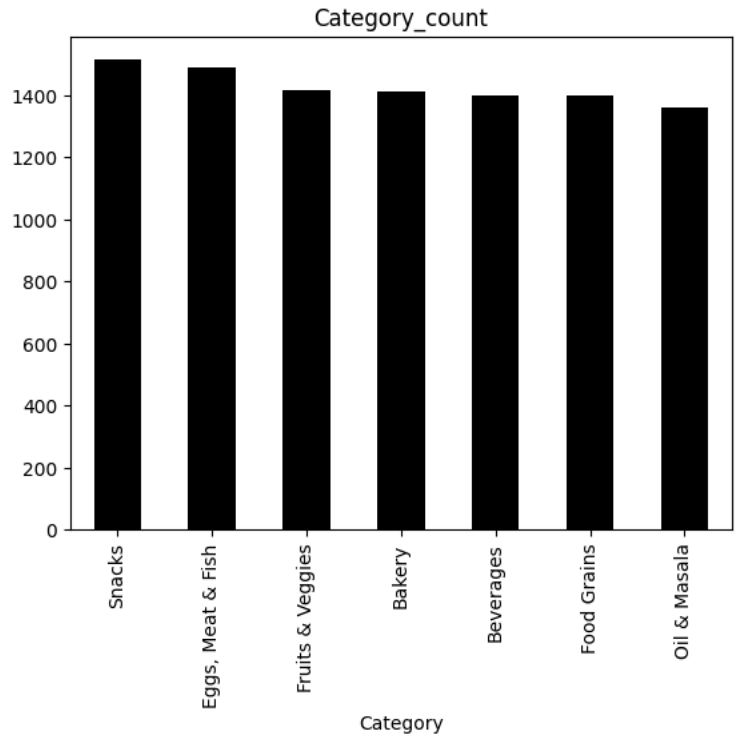
Day	Year	Profit
Friday	**2015**	18634.51
	2016	9325.73
	2017	171111.61
	2018	208060.83
Monday	**2015**	112745.40
	2016	119350.68
	2017	161272.15
	2018	212213.49
Saturday	**2015**	136161.56
	2016	162703.69
	2017	160171.75
	2018	226858.49
Sunday	**2015**	125948.86
	2016	143804.63
	2017	157018.25
	2018	227777.84
Thursday	**2015**	79110.52
	2016	71754.57
	2017	27237.69
	2018	31428.09
Tuesday	**2015**	148958.17
	2016	158309.76
	2017	173654.87
	2018	235442.58
Wednesday	**2015**	130970.09
	2016	131943.93
	2017	102749.90
	2018	102401.56

dtype: float64

Data Visualization

```
# pandas
# matplotlib.pyplot
# seaborn
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns


# Bar chart
# histogram
# line chart
# scatter plot
# pie chart
# Boxplot


s['Category'].value_counts().plot(kind = 'bar', title= 'Category_count', color = 'black')
```

```
<Axes: title={'center': 'Category_count'}, xlabel='Category'>
```



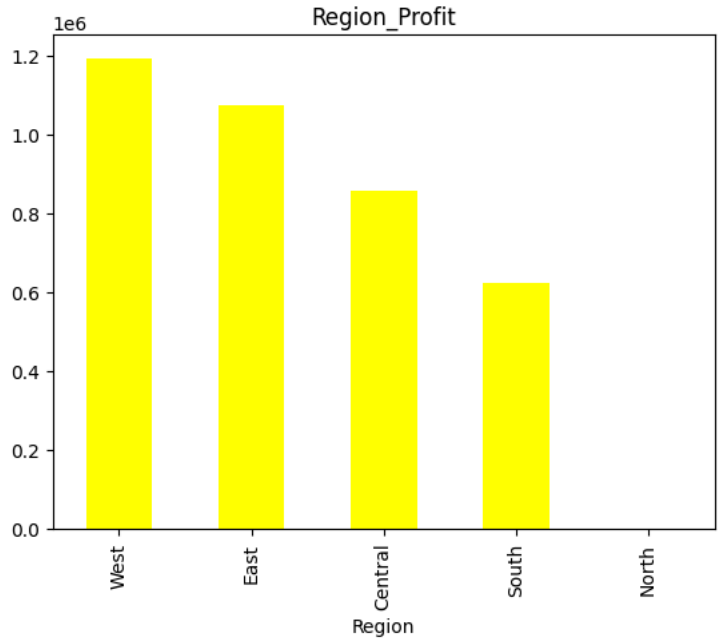

```
s.groupby('Region')['Profit'].sum().sort_values(ascending = False)
```

Region	Profit
West	1192004.61
East	1074345.58
Central	856806.84
South	623562.89
North	401.28

dtype: float64

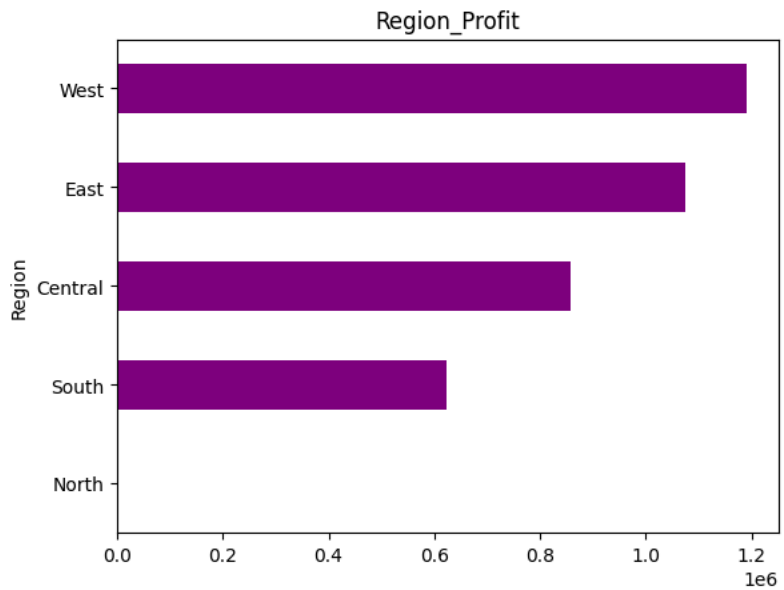
```
s.groupby('Region')['Profit'].sum().sort_values(ascending = False).plot(kind = 'bar', title = 'Region_Profit', color = 'yellow')
```

```
<Axes: title={'center': 'Region_Profit'}, xlabel='Region'>
```



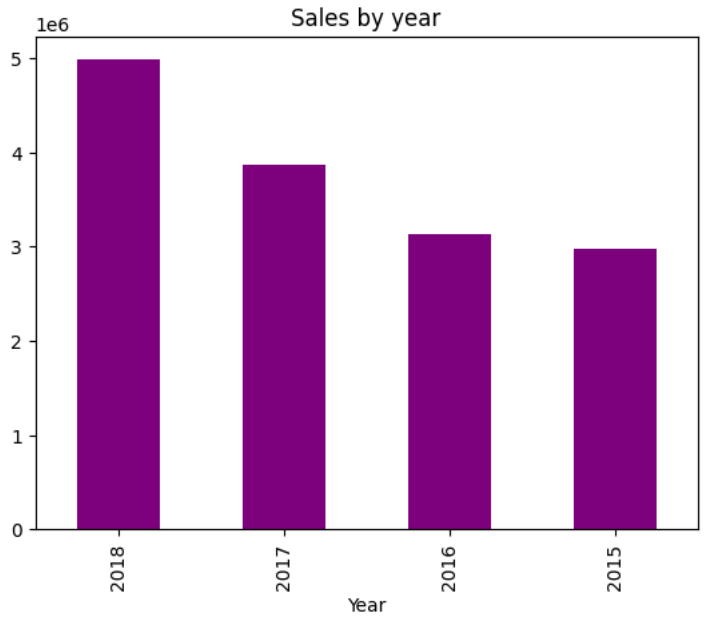

```
s.groupby('Region')['Profit'].sum().sort_values().plot(kind = 'barh', title = 'Region_Profit', color = 'purple')
```

```
<Axes: title={'center': 'Region_Profit'}, ylabel='Region'>
```




```
s.groupby('Year')['Sales'].sum().sort_values(ascending = False).plot(kind = 'bar', title = 'Sales by year', color = 'purple')
```

```
<Axes: title={'center': 'Sales by year'}, xlabel='Year'>
```




```
rp = s.groupby('Region')['Profit'].sum().sort_values(ascending = False)
```

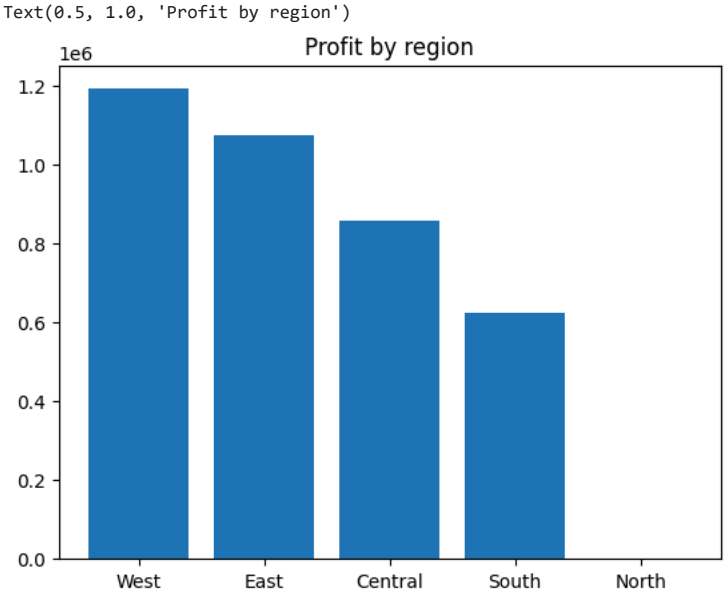
```
rp
```

Region	Profit
West	1192004.61
East	1074345.58
Central	856806.84
South	623562.89
North	401.28

dtype: float64

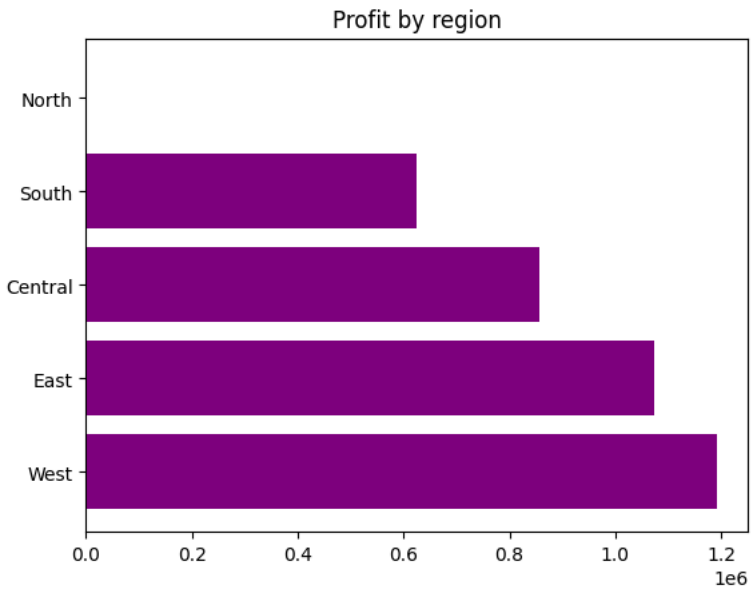
```
plt.bar(rp.index, rp.values)
plt.title('Profit by region')
# plt.xlabel('Region')
# plt.ylabel('Profit')
```

```
Text(0.5, 1.0, 'Profit by region')
```



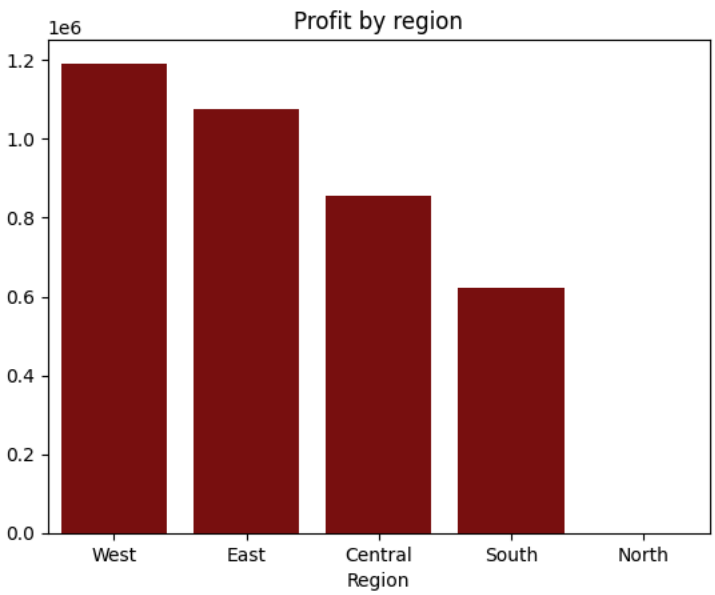
```
plt.barh(rp.index, rp.values, color = 'purple')
plt.title('Profit by region')
```

```
Text(0.5, 1.0, 'Profit by region')
```



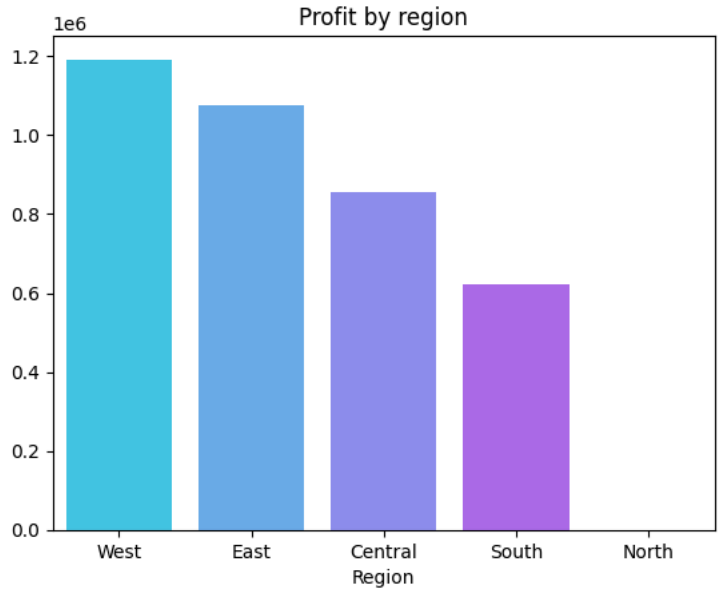

```
sns.barplot(x = rp.index, y = rp.values, color= 'darkred')
plt.title('Profit by region')
```

```
Text(0.5, 1.0, 'Profit by region')
```



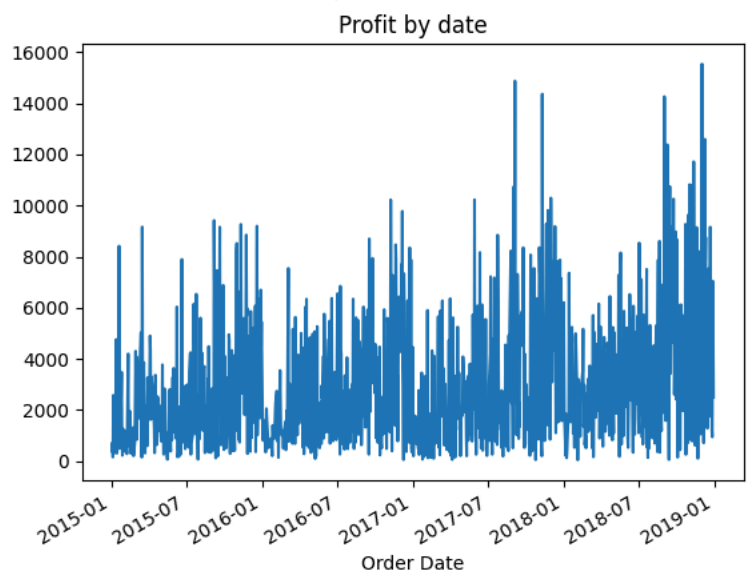

```
sns.barplot(x = rp.index, y = rp.values,  palette = 'cool')
plt.title('Profit by region')
```

```
/tmp/ipython-input-135-641165821.py:1: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend

  sns.barplot(x = rp.index, y = rp.values,  palette = 'cool')
Text(0.5, 1.0, 'Profit by region')
```



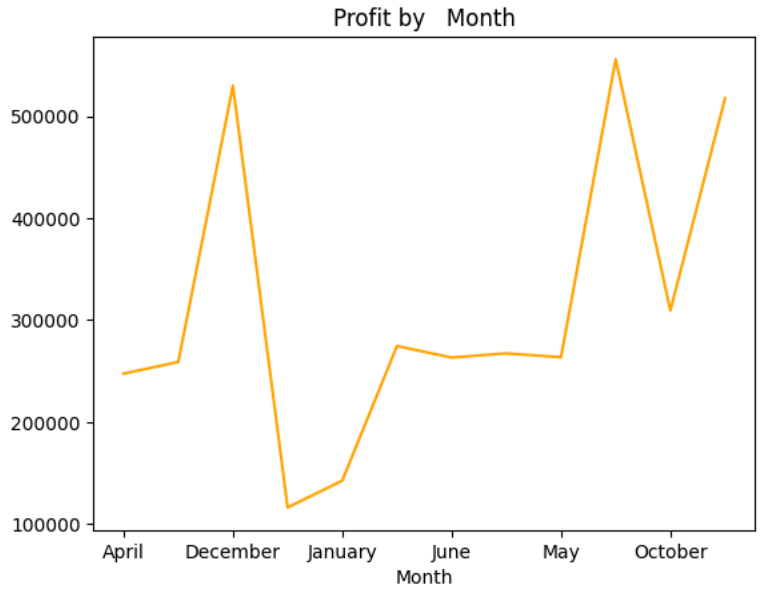

```
# Line chart
s.groupby('Order Date')['Profit'].sum().plot(kind = 'line', title = 'Profit by date')
```

```
<Axes: title={'center': 'Profit by date'}, xlabel='Order Date'>
```



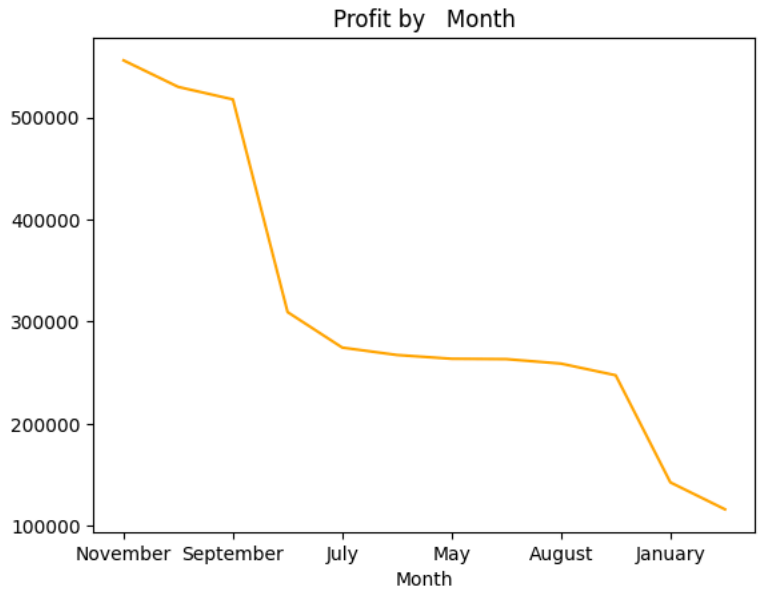

```
s.groupby('Month')['Profit'].sum().plot(kind = 'line', title = 'Profit by   Month', color = 'orange')
```

```
<Axes: title={'center': 'Profit by   Month'}, xlabel='Month'>
```




```
s.groupby('Month')['Profit'].sum().sort_values(ascending = False).plot(kind = 'line', title = 'Profit by   Month', color = 'orange')
```

```
<Axes: title={'center': 'Profit by   Month'}, xlabel='Month'>
```



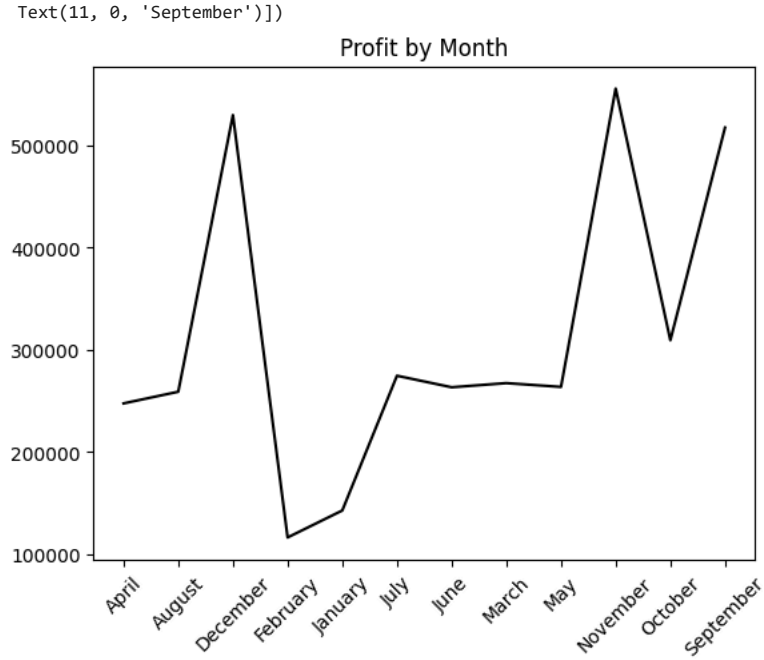

```
pm = s.groupby('Month')['Profit'].sum()
pm
```

Month	Profit
April	247476.97
August	258912.09
December	530036.44
February	116082.21
January	142518.52
July	274594.57
June	263296.16
March	267347.33
May	263643.40
November	556048.05
October	309376.90
September	517788.56

dtype: float64

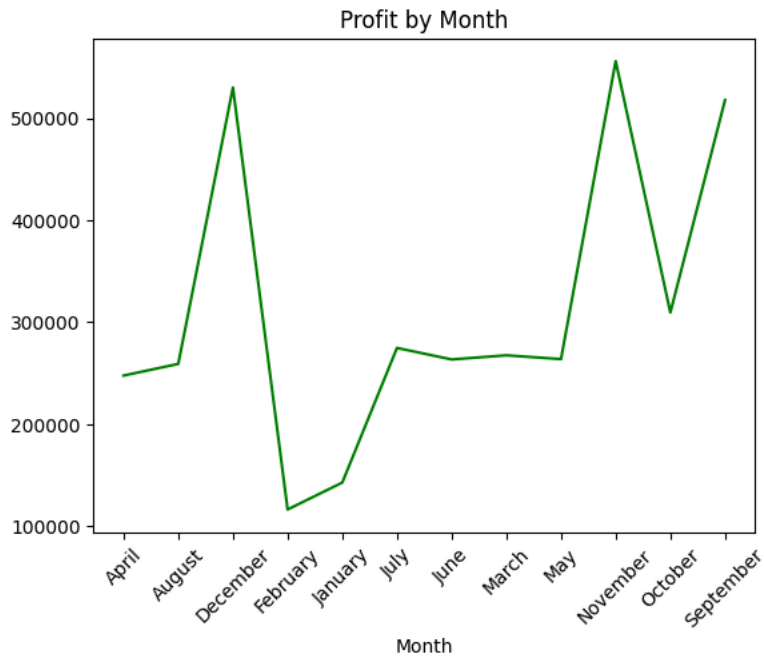
```
plt.plot(pm.index, pm.values, color = 'black')
plt.title('Profit by Month')
plt.xticks(rotation = 45)
```

```
([0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11],
 [Text(0, 0, 'April'),
  Text(1, 0, 'August'),
  Text(2, 0, 'December'),
  Text(3, 0, 'February'),
  Text(4, 0, 'January'),
  Text(5, 0, 'July'),
  Text(6, 0, 'June'),
  Text(7, 0, 'March'),
  Text(8, 0, 'May'),
  Text(9, 0, 'November'),
  Text(10, 0, 'October'),
  Text(11, 0, 'September')])
```



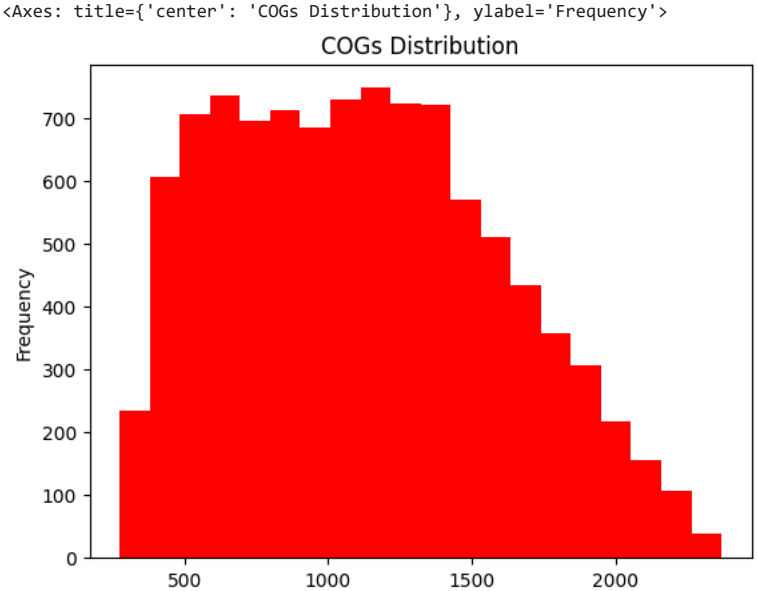

```
sns.lineplot(x= pm.index, y = pm.values, color = 'green')
plt.title('Profit by Month')
plt.xticks(rotation = 45)
```

```
([0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11],
 [Text(0, 0, 'April'),
  Text(1, 0, 'August'),
  Text(2, 0, 'December'),
  Text(3, 0, 'February'),
  Text(4, 0, 'January'),
  Text(5, 0, 'July'),
  Text(6, 0, 'June'),
  Text(7, 0, 'March'),
  Text(8, 0, 'May'),
  Text(9, 0, 'November'),
  Text(10, 0, 'October'),
  Text(11, 0, 'September')])
```



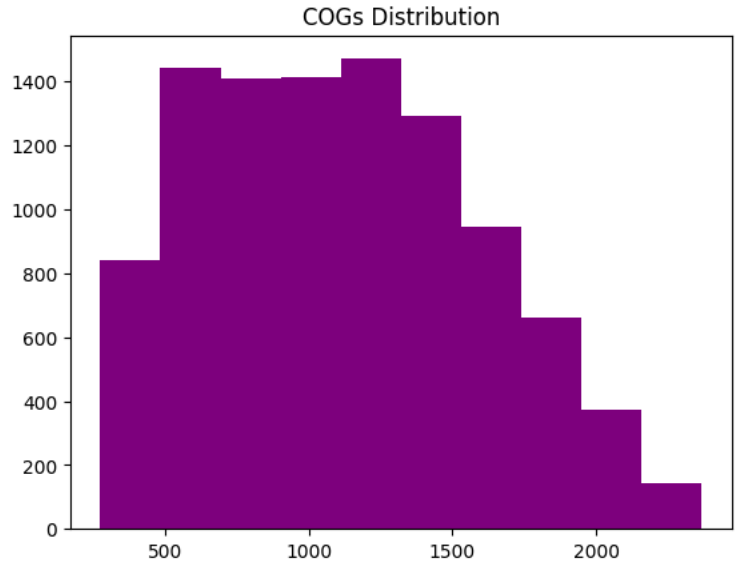

```
s['COGs'].plot(kind = 'hist', title = 'COGs Distribution', bins = 20, color = 'red')
```

```
<Axes: title={'center': 'COGs Distribution'}, ylabel='Frequency'>
```



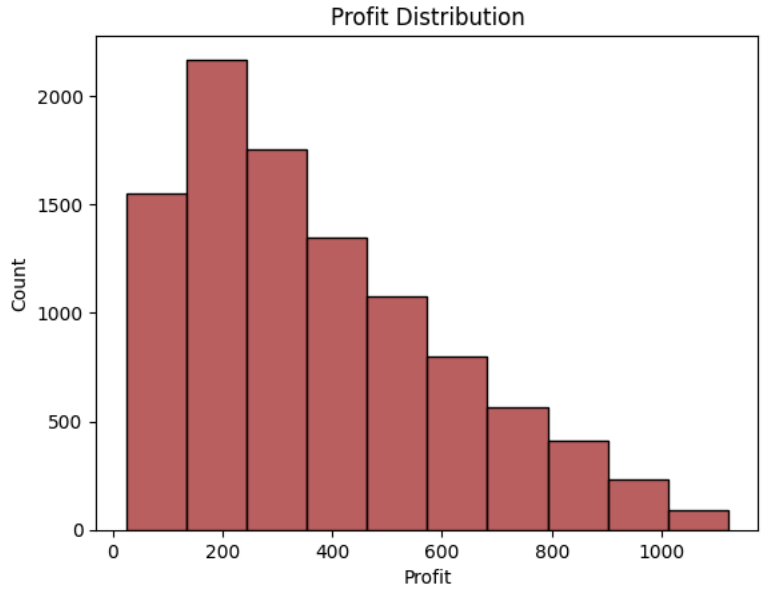

```
plt.hist(s['COGs'], color = 'purple')
plt.title('COGs Distribution')
```

```
Text(0.5, 1.0, 'COGs Distribution')
```




```
sns.histplot(s['Profit'], color = 'brown', bins = 10)
plt.title('Profit Distribution')
```

```
Text(0.5, 1.0, 'Profit Distribution')
```




```
s['Profit']
```

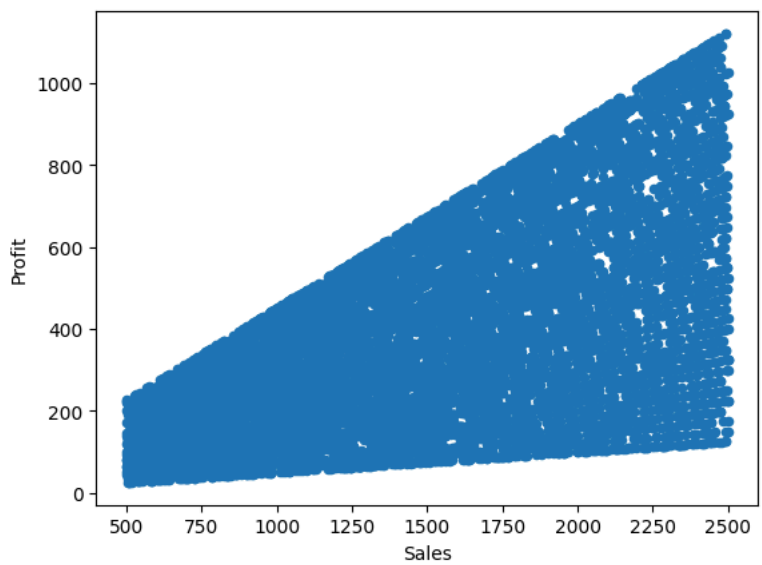
	Profit
0	401.28
1	149.80
2	165.20
3	89.60
4	918.45
...	...
9989	359.10
9990	71.70
9991	501.44
9992	597.24
9993	165.44

9994 rows × 1 columns

dtype: float64

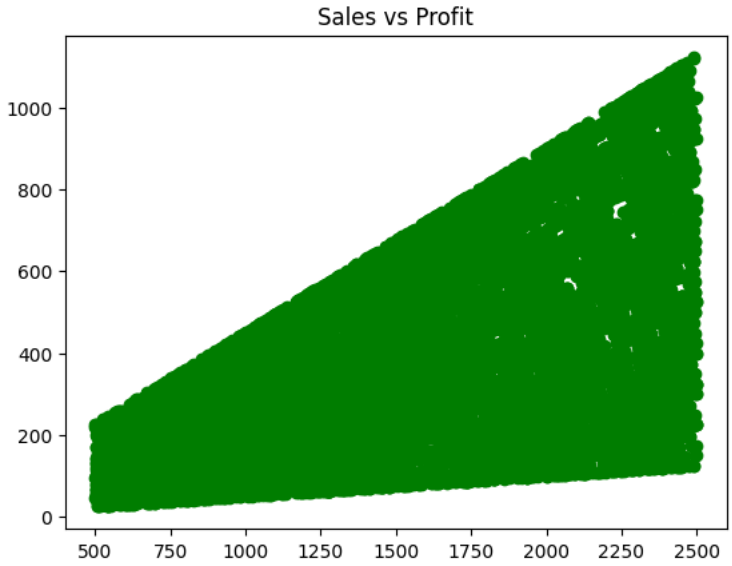
```
s.plot(kind = 'scatter', x = 'Sales', y = 'Profit')
```

```
<Axes: xlabel='Sales', ylabel='Profit'>
```




```
plt.scatter(s['Sales'], s['Profit'], color = 'green')
plt.title('Sales vs Profit')
```

```
Text(0.5, 1.0, 'Sales vs Profit')
```

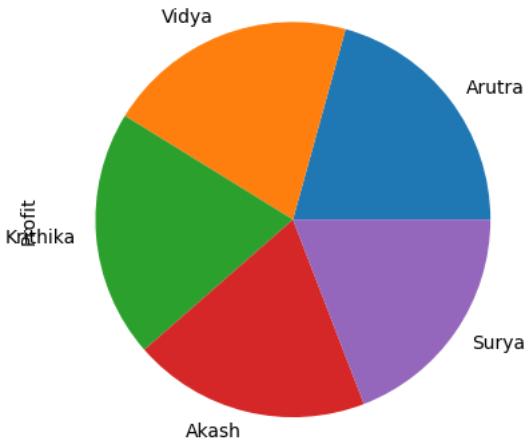



```
sns.scatterplot(x = 'Sales', y = 'Profit', data = s)
plt.title('Sales vs Profit')
```

```
Text(0.5, 1.0, 'Sales vs Profit')
```

```
# pie chart
s.groupby('Customer Name')['Profit'].sum().sort_values(ascending = False).head(5).plot(kind = 'pie')
```

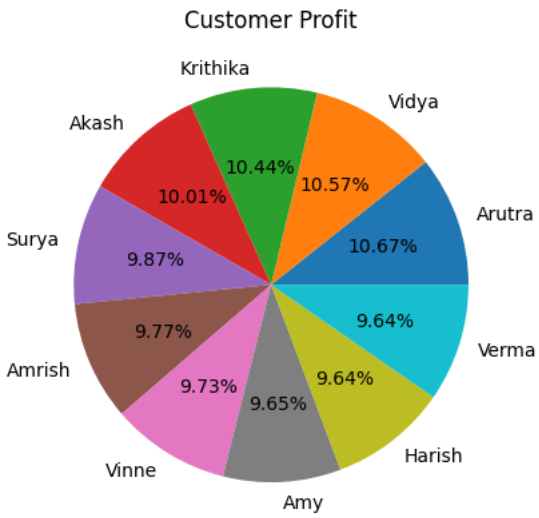
```
<Axes: ylabel='Profit'>
```




```
cp = s.groupby('Customer Name')['Profit'].sum().sort_values(ascending = False).head(10)
```

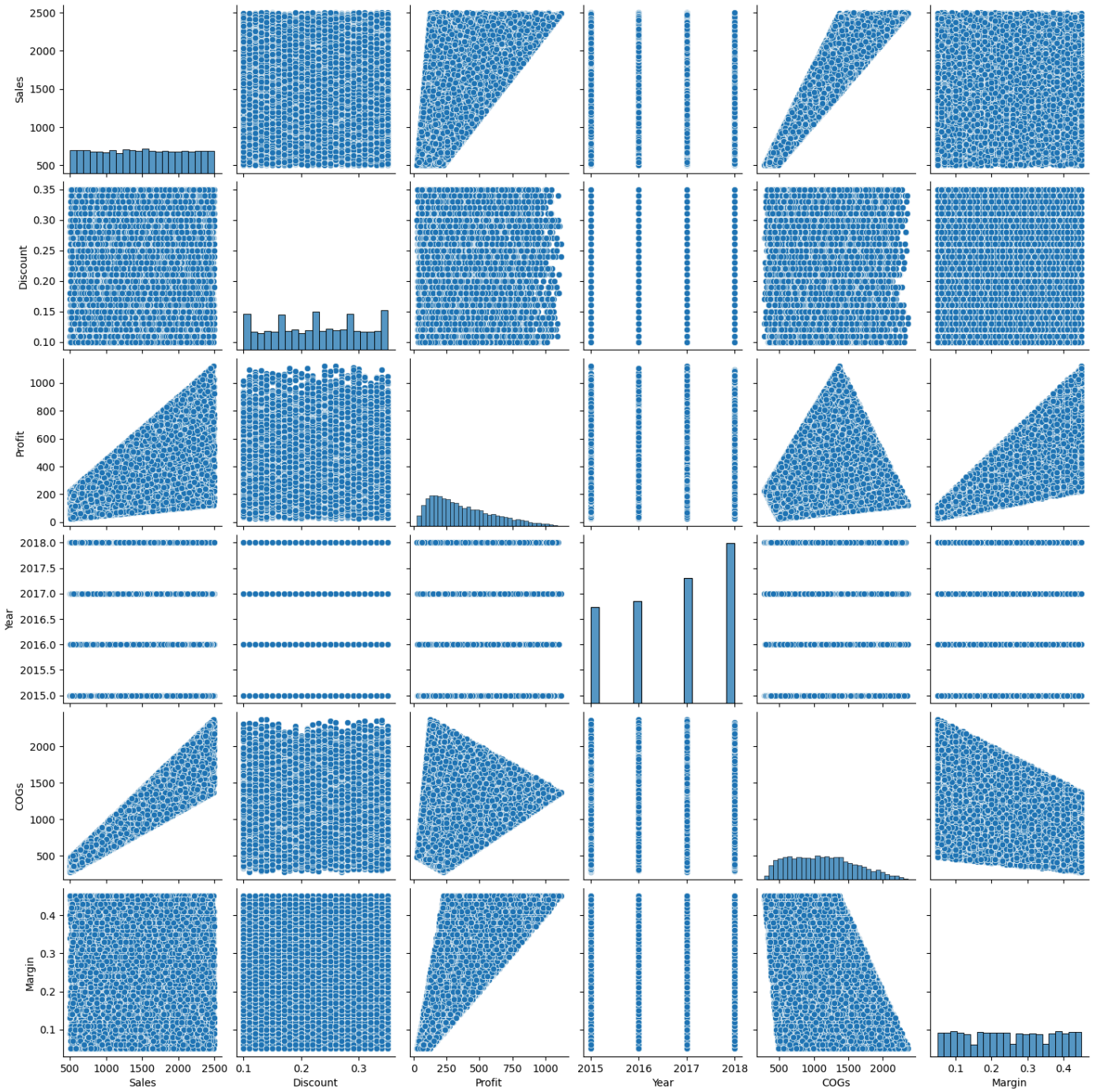
```
plt.pie(cp, labels = cp.index, autopct = '%1.2f%%')
plt.title('Customer Profit')
```

```
Text(0.5, 1.0, 'Customer Profit')
```



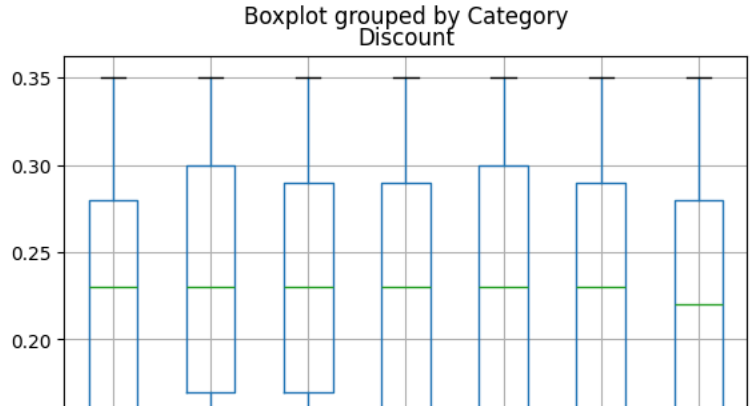

```
sns.pairplot(s)
```

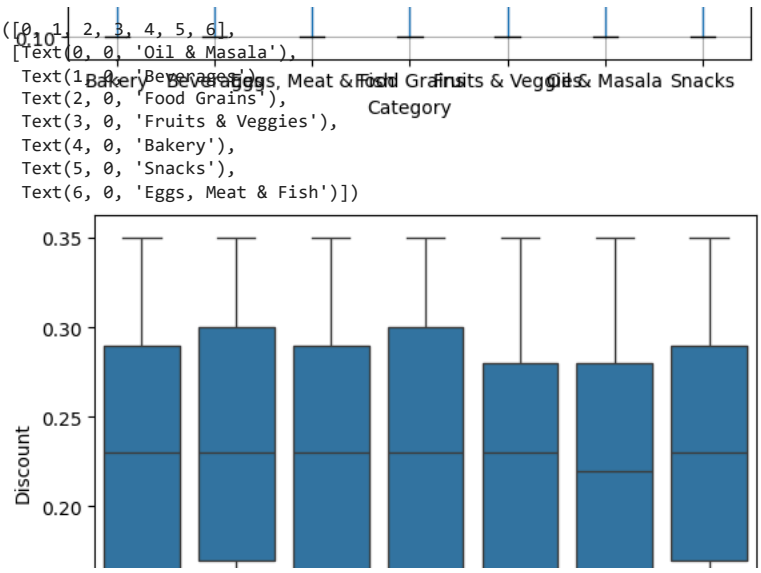
```
<seaborn.axisgrid.PairGrid at 0x79568df0ce50>
```




```
s.boxplot(column = 'Discount', by = 'Category')
```

```
<Axes: title={'center': 'Discount'}, xlabel='Category'>
```




```
sns.boxplot(x = 'Category', y = 'Discount', data = s)
plt.xticks(rotation = 45)
```

```
([0, 1, 2, 3, 4, 5, 6],
 [Text(0, 0, 'Oil & Masala'),
  Text(1, 0, 'Beverages'),
  Text(2, 0, 'Food Grains'),
  Text(3, 0, 'Fruits & Veggies'),
  Text(4, 0, 'Bakery'),
  Text(5, 0, 'Snacks'),
  Text(6, 0, 'Eggs, Meat & Fish')])
```